

Solution Code - Spring Core and Maven - Constructor_Dependency_Injection

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: src\main\java\com\library\BookRepository.java

```
1 | package com.library;
2 |
3 | public class BookRepository {
4 |     public String getDbInfo() { return "Connected to Database Catalog"; }
5 | }
```

Source File: src\main\java\com\library\BookService.java

```
1 | package com.library;
2 |
3 | public class BookService {
4 |     private BookRepository repository;
5 |     public BookService(BookRepository repo) {
6 |         this.repository = repo;
7 |     }
8 |     public void showDetails() {
9 |         System.out.println("BookService Active: " + repository.getDbInfo());
10 |    }
11 | }
```

Source File: src\main\java\com\library\LibraryApp.java

```
1 | package com.library;
2 |
3 | import org.springframework.context.ApplicationContext;
4 | import org.springframework.context.support.ClassPathXmlApplicationContext;
5 |
6 | public class LibraryApp {
7 |     public static void main(String[] args) {
8 |         ApplicationContext context = new ClassPathXmlApplicationContext("applicationContext.xml");
9 |         BookService service = (BookService) context.getBean("bookService");
10 |         service.showDetails();
11 |     }
12 | }
```

Source File: src\main\resources\applicationContext.xml

```
1 | <beans xmlns="http://www.springframework.org/schema/beans"
2 |       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3 |       xsi:schemaLocation="http://www.springframework.org/schema/beans
4 |       http://www.springframework.org/schema/beans/spring-beans.xsd">
5 |
```

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```
6 |      <bean id="bookRepository" class="com.library.BookRepository" />
7 |
8 |      <bean id="bookService" class="com.library.BookService">
9 |          <constructor-arg ref="bookRepository" />
10 |      </bean>
11 | </beans>
```